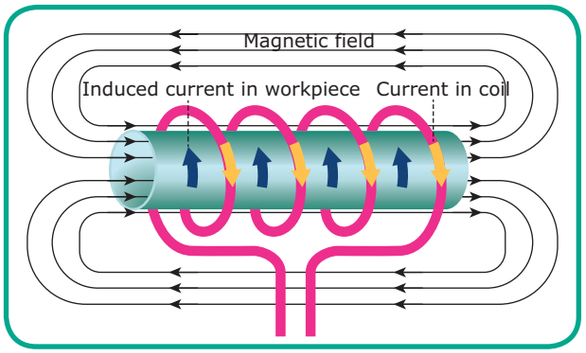


Figure 2.6 Electro Magnetic Force

2.2.3 Types of Inductor

Inductor can be divided in two categories.

1. Fixed inductors
2. Variable inductors

2.2.3.1 Fixed Inductor

The symbol of fixed inductor is shown in Figure 2.7 and can be further divided into 3 categories depending on the type of cores.

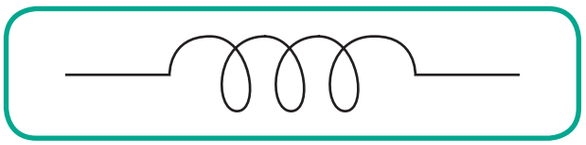


Figure 2.7 Fixed Inductor

They are,

1. Air Core Inductors
2. Iron Core Inductors
3. Ferrite Core Inductors

Air Core Inductors: Air core inductors consist of a few turns of wire wound on a hollow former. In radio frequency applications, where a very low value of inductance is required, air core inductors are generally used. It is shown in Figure 2.8.

Iron Core Inductors: In this, the former is generally made up of silicon steel in the form of thin laminated sheets. These are used in low frequency applications such as filter circuits in power

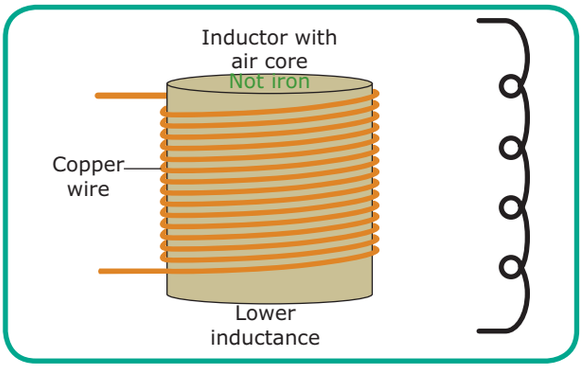


Figure 2.8 Air Core Inductors

supplies, chokes in fluorescent tubes. The iron core inductor is shown in Figure 2.9.

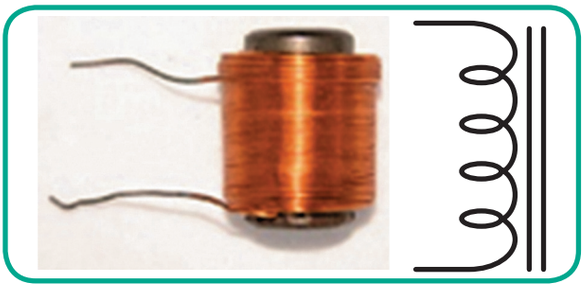


Figure 2.9 Iron Core Inductors

Ferrite Core Inductors: Ferrite core inductors consist of a few turns of wire wound on a ferrite materials. A ferrite is basically an insulator having very high permeability. Ferrite core inductor is suitable for high frequency applications and is shown in Figure 2.10.

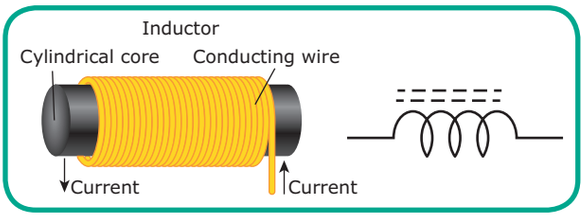


Figure 2.10 Ferrite Core Inductors

2.2.3.2. Variable inductors

In certain application such as tuned circuits, it is required to vary the inductance from minimum value to a maximum value. Ferrite core variable inductors are generally used for this purpose. The variable inductors are shown in Figure 2.11.

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